A Century of Falling Costs? Wright's Law, Commodity Prices, and the Limits of Learning-by-Doing, 1925–2026

Kilambi Research

September 2026

Kilambi Research Working Paper No. KR-WP-2026-05

Abstract

We assemble a cross-sector longitudinal dataset of 6,662 annual observations on 98 manufactured technologies and commodity price series from 57 sources, covering 1774–2026 in raw span. All 2026 partial-year observations are excluded from estimation, so 2025 is the latest fitted year; deflated-price estimation begins in 1913 where deflators exist. We estimate the real cost dynamics of technologies and the price dynamics of a 55-series commodity-and-context basket over the last century. Estimating Wright's Law, $\ln C = a + b \ln Q$, on the six technologies with usable cumulative production data, we find learning rates of 38.6% [37.1, 40.1] for DRAM (1971–2007), 32.5% [31.9, 33.0] for computers (1967–2025), 27.7% [25.7, 29.7] for solar PV modules (2000–2024), 8.4% [6.6, 10.2] for US aluminum smelting energy intensity (1960–2003), and no detectable learning for US wind installed cost (0.6% [−2.7, 3.9], 1998–2023) or US wind turbine prices (−5.3% [−12.5, 1.3], 1998–2016). Across 31 manufactured-goods time trends the median real rate of change is −4.05% per year, against +0.51% per year across the 50 deflated commodity price trends (+0.53%/yr for the full 55-series commodity-and-context basket, which includes five nominal index series never deflated): technologies fall, commodities drift and cycle. Counterexamples are real and instructive: US construction costs rose 4-fold in real terms 1913–2010 (+1.6%/yr), nuclear construction costs rose with cumulative experience in several countries (published literature only; no reactor-level series exists in this dataset), and hedonic quality-adjusted indices embed the producer's quality model, so their "learning rates" are model output. Because cumulative production and calendar time are nearly collinear (variance inflation factors 5.9–60.9), we do not interpret bivariate experience-plus-time specifications structurally. Every quantitative claim in this paper is traceable to a results file, a data row, a figure, or a cited publication.

Keywords: Wright's Law; learning curves; experience curves; technological change; commodity prices; hedonic quality adjustment; energy transitions.
JEL: O33, O47, Q41, N70.

Contents

1 Research question and hypotheses **3**

2 Literature review **3**

3 Data and sources **5**

4 Methodology **6**

5 Normalization and quality-adjustment framework **7**

6 Statistical models **8**

7 Results **8**

8 Commodities and the long-run price environment **9**

9 Counterexamples: when costs did not fall **10**

10 Interpretation **10**

11 Threats to validity **11**

12 Conclusion **12**

13 Audit protocol and reproducibility **12**

A Data provenance register **16**

B Additional estimates **17**

1 Research question and hypotheses

How has the real cost of making things changed over the last century, and what drives the differences across sectors? The popular narrative is one of relentless technological deflation: computing costs fall orders of magnitude per decade, solar modules follow "Swanson's law," and each generation of technology is cheaper than the last. The commodity record tells a different story: one of booms, busts, and long flat stretches punctuated by geopolitical shocks. This paper asks whether the deflation narrative survives contact with a defensible, cross-sector, century-scale dataset, and where it breaks down.

We test five hypotheses:

H1. For manufactured technologies, real unit cost declines log-linearly in cumulative production (Wright's Law), with a stable, estimable learning rate.

H2. Learning rates differ systematically across technology classes: fastest in information technologies, slower in mature process industries, and absent where input costs dominate.

H3. Extractive and agricultural commodity prices do not "learn": their long-run real trends are small relative to their volatility, and best described as drift plus cycles, not experience curves.

H4. Because cumulative production typically grows exponentially, a pure time trend approximates Wright's Law closely in fit (the Sahal indistinguishability argument), so the two specifications must be compared on identical samples rather than assumed distinct.

H5. Learning fails or reverses in identifiable cases: construction, nuclear power, and input-cost-exposed equipment show zero or negative measured learning.

The contribution is empirical discipline. We build one harmonized dataset (Section 3), apply one documented deflator policy (Section 5), estimate with one reproducible script using heteroskedasticity-robust inference (Section 4), and report what the data support, including the null results (Sections 7–9). A claim index maps every quantitative statement to its source. This is a retrospective study: every estimate describes the historical sample through 2025; none is a forecast of future costs.

The remainder of the paper is organized as follows. Section 2 reviews the experience-curve and commodity-price literatures. Section 3 describes the dataset and the unified evidence grading; Sections 4–6 the estimation, normalization, and statistical models. Section 7 reports the learning-rate and time-trend results, Section 8 the commodity price environment, and Section 9 the counterexamples. Section 10 interprets, Section 11 states the threats to validity and the conditions that would falsify the results, and Section 12 concludes. Tables follow the conclusion; the audit protocol and data provenance register are appended.

2 Literature review

The experience curve. Wright (1936) observed that direct labor hours per airframe at Curtiss-Wright fell by a roughly constant fraction (about 20%, an 80% progress ratio) with each doubling of cumulative output. In the project convention used throughout this paper, the experience curve is estimated as $\ln C = a + b \ln Q$, so that $b < 0$ measures the cost elasticity with respect to cumulative production Q, and the learning rate $LR = 1 - 2^b$ is the fractional cost reduction per doubling of cumulative production. Arrow (1962) endogenized the mechanism: productivity growth arises as a by-product of cumulative gross investment ("learning by doing") providing the theoretical basis for treating experience, rather than calendar time, as the driver of cost decline. Thompson (2012) surveys the organizational-learning evidence and the identification problems that attend it.

Moore's Law and time-based alternatives. Moore (1965) projected that components per integrated circuit would double yearly (later revised to every two years), a time-exponential regularity that Nagy et al. (2013) generalized as "Moore's Law" for any technology: cost falls exponentially in calendar time. Using 62 technologies, Nagy et al. (2013) hindcast six candidate laws and found Wright's Law the best forecaster, with Moore's Law close behind, and showed why: production itself tends to grow exponentially, so that Sahal (1979)'s indistinguishability argument makes the two laws nearly equivalent in practice. Forecast error grows with horizon at roughly 2.5% per year in log terms (Nagy et al. 2013). Farmer and Lafond (2016) extended the framework, showing technological progress is predictable enough to discipline climate-policy modeling.

Energy-technology learning rates. The energy literature has produced the densest set of estimates. McDonald and Schrattenholzer (2001) compiled early learning rates for energy technologies; Neij (1997) applied experience curves to renewables diffusion; Junginger et al. (2005) estimated global wind-farm experience curves; Nemet (2006) and Nemet (2009) decomposed PV cost reductions beyond the learning curve, separating scale, R&D, and input-price effects. Smil (2010) argues from the long history of energy systems that full-scale transitions are inherently slow and prolonged, cautioning against extrapolating fast-learning niches to whole systems. Rubin et al. (2015) reviewed reported learning rates for 11 electricity supply technologies and found order-of-magnitude variation across studies, unexplained by time window, region, or specification choices. Yeh and Rubin (2012) catalogued the uncertainties in technology experience curves. Ferioli et al. (2009) warned that single-factor curves misattribute cost change and proposed component-level learning. For storage, Schmidt et al. (2017) constructed experience curves for 11 electrical storage technologies, and Ziegler and Trancik (2021) re-examined lithium-ion improvement rates, finding roughly 20% per doubling for all cell types (24% for cylindrical cells) over 1992–2016. Nordhaus (2014) cautioned against naïve use of learning models for endogenous technical change in integrated assessment.

Negative learning: nuclear. The canonical counterexample is nuclear power. Lovering et al. (2016) harmonized overnight construction costs for 349 reactors across seven countries and found costs rising with cumulative experience in several countries (notably the United States and France) while falling in South Korea. Grübler (2010) documented the French case as "negative learning by doing," and Koomey and Hultman (2007) analyzed busbar costs for 99 US reactors (1970–2005). We discuss this literature in Section 9; no reactor-level series could be traced to a primary source for our dataset, so nuclear enters this paper as published literature only.

Quality adjustment and the measurement of cost decline. Long-run cost series are inseparable from quality measurement. Gordon (1990) built quality-adjusted price indexes for durable goods from Sears catalogs; Raff and Trajtenberg (1995) pushed hedonic automobile price measurement back to 1906–1940, finding hedonic prices fell about 5% per year; Gordon's commercial-aircraft indexes (Gordon 1990, ch. 4) fall roughly 7% per year while the official aircraft PPI rose: a wedge that is the point, not a bug, of quality adjustment. Nordhaus (1996) showed that a century of lighting (an energy *service* whose technology changed from tallow to gas to electric) defies product-based cost measurement entirely; Fouquet and Pearson (2006) priced UK lighting services over seven centuries. Nordhaus (2007) reconstructed two centuries of computing cost as a technological-frontier series.

Commodities. Commodity prices are driven by geology, inventories, market structure, and geopolitics, not by cumulative production. Jacks (2019) developed a typology of real commodity prices in the long run (boom-bust cycles rather than trends). Hummels (2007) documented the

postwar collapse of air-freight costs (about −8%/yr 1955–72, −3.5%/yr 1972–2003) alongside a structural break in the early 1970s. Our commodity analysis follows this tradition: trends are descriptive, volatility and persistence are the objects of interest, and no learning curve is ever fit to a commodity price.

3 Data and sources

Corpus. The master dataset (`data/master_costs.csv`) contains 6,662 annual observations on 98 technologies and commodity price series drawn from 57 registered sources. Raw coverage spans 1774–2026; all 2026 partial-year observations are excluded from every fit, so 2025 is the latest fitted year. Deflation follows the project policy of Section 5: our own deflated-price estimation begins in 1913, where the PPI-AC deflator exists, while a few series arrive already deflated by their sources (e.g., Shiller real home prices) and context series such as the MeasuringWorth wage index begin earlier. Five sector builders contributed: computing/telecom (439 rows), energy technologies and fuels (975), macro deflators and context series (642), materials and commodities (4,078), and transport/industry (528). Every row carries a stable identifier, a source key, a data-status tag, a quality grade, and a methodology note; every row that enters deflated-price estimation carries a deflator record: 1,332 commodity rows (nominal or production-only series that never enter a deflated fit, 969 of them dated 1913 or later) carry no deflator. Every transformation from raw to master is logged in `data/transform_log.csv`. Raw files are never overwritten. Table 1 summarizes the estimation samples; Table 4 the evidence-grading composition.

Unified evidence grading. Each observation carries a two-part evidence tag: a *data status* and a *source-quality grade*. Data status is **Observed** (5,706 rows: values taken directly from a source as published, without transformation), **Reconstructed** (675 rows: values assembled from primary fragments or re-tabulated by stated methods, e.g., Fouquet–Pearson lighting benchmarks, McCallum disk prices, Shiller house prices), **Modelled** (281 rows: statistical model output, principally hedonic quality-adjusted indexes such as BEA computers, BLS semiconductor PPIs, Raff–Trajtenberg autos, and Gordon aircraft), or **Assumed** (0 rows in the master file: parameters fixed by the analyst rather than measured; none enter estimation). Gaps are left as gaps; no backfilling is performed anywhere. Source quality is **A** (4,000 rows: official primary producer with documented, consistent methodology), **B** (2,624 rows: reputable secondary producer or peer-reviewed reconstruction with known breaks), or **C** (38 rows: provisional or fragmentary; excluded from all headline fits). The tags combine: a BEA computers row is Modelled/A, a McCallum disk-price year is Reconstructed/B, a BLS PPI row is Observed/A, and an analytical scenario parameter would be Assumed/C. Headline estimates use grades A and B only; grade C is excluded.

Key series. Computing: NHGRI cost per human genome (2001–2022, A); BEA hedonic computers-and-peripherals price index and real investment (1959–2025, A, modelled); BLS semiconductor PPIs (1981–2025, A, modelled); FCC telephone rates (1930–2007); McCallum's annual-minimum hard-disk $/GB (1956–2017, reconstructed); Performance Curve Database DRAM $/bit with shipments (1971–2007, reconstructed). Energy: OWID solar PV module $/W (1975–2024, harmonizing Nemet, Farmer–Lafond, and IRENA); LBNL Wind Report 2024 US turbine-order prices (1997–2016), installed costs (1983–2023), and PPA prices (1998–2022); BNEF lithium-ion pack $/kWh (2010–2024); OWID crude oil (nominal 1861–2025, real from 1913); World Bank Pink Sheet fuels (1960–2025); Fouquet–Pearson UK lighting service price (three benchmark years: 1900, 1950, 2000). Materials: USGS Data Series 140 unit values for eight metals plus cement (1900–2021, US); World Bank Pink Sheet global metals, grains, and

fertilizer prices (1960–2025); IMF commodity prices via FRED (1992–2026); BLS commodity-group PPIs (1926–2025); USDA wheat farmgate price (1866–2026); DOE aluminum smelting energy intensity (1960–2003) with cumulative US primary production. Transport/industry: Raff–Trajtenberg quality-adjusted autos (1906–1983, with a 1941–46 WWII gap); Gordon quality-adjusted commercial aircraft (1957–1983); Hummels digitized air-freight revenue per tonne-km (1955–2002, reconstructed); BTS freight revenue per ton-mile (air and rail 1960–2014; truck 1990–2007); BLS motor-vehicle, machinery, and aircraft-manufacturing PPIs (1937/1939/1985–2025). Context and counterexample series: ENR Construction Cost Index (1908–2010), Shiller US real house-price index (1890–2025), MeasuringWorth manufacturing compensation (1790–2025) and unskilled-wage index (1774–1940).

What is not in the dataset. Several priority targets could not be admitted without fabrication and are documented as excluded: the Ziegler–Trancik lithium-ion series (Harvard Dataverse download returned HTTP 403; BNEF pack survey and DOE cell endpoints included instead, kept strictly separate); the Lovering–Yip–Nordhaus reactor-level nuclear data (never released as journal supplementary material); Jacks–Stuermer dry-bulk freight rates (no downloadable replication dataset); IAI global aluminum smelting energy (login wall); Smil's iron/steelmaking energy series (book-only); heat pumps, motors, transformers, and transmission (no verified public long-run constant-quality series); industrial robots (single clean price point); pharmaceuticals (product churn defeats century comparability); pre-1990 rare-earth and pre-2015 lithium prices (opaque/paywalled markets). Conflicting sources for the same technology-year are kept as separate rows, never averaged.

4 Methodology

All estimation is performed by `results/estimation.py` (seeded, deterministic; re-running reproduces the four results CSVs byte-identically). All regressions are OLS with robust (HC1) standard errors; 95% confidence intervals use the t_{N-k} critical value, not the normal approximation.

Sample construction. Headline samples use quality grades A and B only; grade C is excluded. All 2026 partial-year rows (19) are excluded from every fit. Minimum N: 8 for learning-curve and time fits, 20 for commodity trends and break tests. Wright samples additionally require finite positive cumulative production Q; the computers series' 1959–1966 rows carry $Q = 0.00$ (a cumsum rounding artifact) and are excluded, so the Wright fit runs 1967–2025. No pooling across `source_id`: the four LBNL wind-turbine-price subseries (US order medians 1997–2016; Vestas, SGRE, Nordex global averages) are estimated separately, never averaged.

Estimators. Wright's Law:

$$\ln C_t = a + b \ln Q_t + \varepsilon_t, \tag{1}$$

with progress ratio $PR = 2^b$ and learning rate $LR = 1 - 2^b$ per doubling of cumulative production. Time trend: $\ln C_t = a + g\,t + \varepsilon_t$, where g is the continuously-compounded annual rate (reported as $100 \times g$, "%/yr," the log-linear approximation; for the fastest series we also report the discrete conversion $\exp(g) - 1$). The bivariate specification $\ln C = a + b \ln Q + g\,t$ is estimated for the six Wright-eligible groups but, as Section 7 shows, b and g are not separately identified. The *time* specification for Wright-eligible groups is estimated on the identical complete-case sample as the Wright fit, so R^2 values are directly comparable. The 25 manufactured groups with no cumulative production data receive time trends only, labeled as such.

Breaks, rolling rates, commodities. Structural breaks are tested with robust (HC1) Wald tests on break dummies $\mathbf{1}(t > \tau)$, $x \cdot \mathbf{1}(t > \tau)$ ($\chi^2(2)$ under the null of no break) at sample midpoints and at documented break years from the build reports (autos 1947, FCC local 1983, BEA 1985, solar 2004/2010, wind CapEx 1998, air freight 1972); sup-Wald scans interior years with ≈15% trimming, with p-values from Andrews (1993) Table I critical values for $q = 2$ by linear interpolation between the tabulated knots (statistics 0–25, p 1.0–0.001). All 23 sup-Wald statistics in this paper exceed the table's largest tabulated statistic (25), so every reported sup-Wald p-value is a crude tail extrapolation, $p \approx 0.001 \times 25/\text{stat}$, not a table interpolation; read them as order-of-magnitude only. Rolling learning rates use 15-calendar-year windows (minimum 10 observations). Commodity trends are log-linear, $\ln P_t = a + g\,t$, on deflated prices; index series are flagged (g is then an index growth rate). Volatility is the standard deviation of consecutive-year log differences; persistence is the AR(1) coefficient on demeaned $\ln P$. Learning curves are *never* fit to commodity prices.

5 Normalization and quality-adjustment framework

The cost metric C. The rule is: use `perf_cost` when `perf_unit` is a true physical denominator ($/W, $/kWh, $/GB, $/bit, $/genome, $/kW, $/tonne-km, $/min, $/access-line-month); use `real_cost` for hedonic or official quality-adjusted indexes. In this dataset `perf_cost` is numerically identical to `real_cost` wherever both exist (maximum relative difference 0.0 across 1,637 rows, verified in-script), so the choice is a documented convention, not a numeric one. We call this choice the **C-metric** and report it per estimate (Table 2).

Deflator policy. The default for US manufactured technology is the Producer Price Index for All Commodities (PPIACO, 1913–, producer basis, full span); WPU11 machinery & equipment for machinery sectors from 1939. Commodities use PPIACO or the matching commodity-group PPI. Non-US nominal series use the country's own official index (UK lighting: UK CPI rebased via OECD/BoE series). Every row that enters deflated-price estimation states its deflator; the base year is 2020 throughout. CPI-U is not the primary deflator. As a contextual check, the build reports compare PPIACO and CPI-U cumulative deflation factors over each manufactured series' span: the largest disagreement is the disk series (CPI-U gives 41% more cumulative inflation 1956–2017), about 0.15 log points on an 8.3 log-point decline, immaterial to learning-rate magnitudes. No CPI-deflated refits were run; the results files report PPI-based estimates only.

Hedonic and modelled values. Seven series are `data_status`=modelled, and their "learning rates" are model output: the BEA computers index (1959–1985 is a reconstruction on the modern hedonic basis, introduced in the December 1985 comprehensive revision); BLS semiconductor PPIs (chain-linked across NAICS reclassifications; US-producer scope misses offshore/fabless mix); Raff–Trajtenberg autos (the 1940–47 postwar splice links hedonic vintages via Griliches/Gordon coefficients; pre-1941 graded A, post-1946 graded B); and Gordon aircraft (already real by construction). We report these estimates because the series are the best available records, but we flag their model-dependence wherever they appear.

Index series and unit values. BLS PPIs, the ENR CCI, the Shiller HPI, and Measuring-Worth wage indices are index series: their "trends" are index growth rates, flagged as such. The ENR CCI is a fixed-weight input-cost index (basket frozen at 1913 quantities; steel pricing basis changed 1996), not a productivity measure. USGS "prices" are unit values of apparent consumption, not quoted market prices. Pink Sheet specifications change at documented dates. Pre-1913 nominal-only rows (USGS, wheat) are excluded from fits; no deflator exists there.

6 Statistical models

Wright's Law. Equation (1) is estimated by OLS on the six technologies with cumulative-production data. The slope b identifies the progress ratio $PR = 2^b$ and the learning rate $LR = 1 - 2^b$, the proportional cost reduction per doubling of Q. Confidence intervals for LR are obtained by mapping the HC1 t_{N-2} slope-CI endpoints through $LR = 1 - 2^b$; because LR is monotone in b, the interval endpoints map directly. We report b with its standard error alongside LR so the transformation is auditable.

Time-trend horserace. For the same six groups we estimate $\ln C_t = a + g\,t$ on the identical complete-case sample and compare R^2. Following Sahal (1979) and Nagy et al. (2013), we expect near-ties whenever Q grows exponentially: if $\ln Q_t \approx c + ht$, then (1) becomes $\ln C_t \approx (a + bc) + bh\,t$, and b and $g = bh$ are the same object viewed through different lenses. The horserace therefore tests fit, not mechanism.

The bivariate trap. Estimating $\ln C = a + b \ln Q + g\,t$ invites a structural reading ("learning net of time") that the data cannot support. Variance inflation factors for $(\ln Q, t)$ range from 5.9 (aluminum) to 60.9 (DRAM), with correlations of 0.91–0.99. We report these coefficients for completeness with an explicit collinearity warning and do not interpret them.

Commodity trends. For each of 55 commodity and context series we estimate $\ln P_t = a + g\,t$ by OLS-HC1. With AR(1) coefficients typically 0.7–1.0, these are near-unit-root processes; g is a descriptive drift, not a structural parameter, and the accompanying volatility (std. of annual log differences) and largest multi-year drawdowns/runups are the economically relevant statistics. Five of the 55 rows are nominal index series (four BLS commodity-group PPIs and the MeasuringWorth unskilled-wage index) estimated without deflation; Section 7 reports the median over the 50 deflated series alongside the full-basket median.

7 Results

Headline learning rates. Table 2 and Figures 2–4 report the six Wright fits. Three technologies learn fast: DRAM at 38.6% [37.1, 40.1] per doubling ($b = -0.704$, SE 0.017, $N = 37$, 1971–2007, $R^2 = 0.984$); computers at 32.5% [31.9, 33.0] ($b = -0.567$, SE 0.006, $N = 59$, 1967–2025, $R^2 = 0.992$); and solar PV modules at 27.7% [25.7, 29.7] ($b = -0.468$, SE 0.020, $N = 25$, 2000–2024, $R^2 = 0.955$). Aluminum smelting energy intensity learns slowly, 8.4% [6.6, 10.2] ($b = -0.127$, SE 0.014, $N = 44$, 1960–2003, $R^2 = 0.858$). Wind shows no detectable learning: installed cost 0.6% [−2.7, 3.9] ($b = -0.009$, SE 0.023, $N = 26$, 1998–2023, $R^2 = 0.004$) and US turbine-order prices −5.3% [−12.5, 1.3] ($b = +0.075$, SE 0.044, $N = 18$, 1998–2016, $R^2 = 0.161$), both confidence intervals cover zero, and turbine prices rose as cumulative capacity grew. The Q proxies are deployment or investment series (solar: cumulative global installed capacity; computers: cumulative real private investment, not depreciated stock; wind: cumulative US capacity; aluminum: cumulative US primary production), so these rates describe the paired series, not pure "experience."

Time trends. The 31 manufactured-goods time trends (Table 5 source rows; Figure 7) span $100 \times g$ from −69.3 to +1.3 per year (continuous log rates). The fastest declines are DNA sequencing ($g = -0.6930$, $N = 22$, 2001–2022, $R^2 = 0.925$; −50.0%/yr discrete), hard-disk storage ($g = -0.4405$, $N = 48$, 1956–2017, $R^2 = 0.928$; −35.6%/yr), DRAM ($g = -0.4354$, $N = 37$, $R^2 = 0.988$; −35.3%/yr), lithium-ion packs ($g = -0.1841$, $N = 15$, 2010–2024, $R^2 = 0.954$; −16.8%/yr), solar PV ($g = -0.1544$, $N = 25$, $R^2 = 0.950$; −14.3%/yr), and computers ($g = -0.1300$, $N = 59$, $R^2 = 0.943$). Mature-sector trends are an order of magnitude

slower: rail freight −1.85%/yr ($N = 31$, $R^2 = 0.823$), motor-vehicle PPI −0.58%/yr ($N = 89$, 1937–2025, $R^2 = 0.646$), machinery PPI −0.35%/yr ($N = 87$, $R^2 = 0.281$), and aircraft-manufacturing PPI +0.39%/yr ($N = 41$, 1985–2025, $R^2 = 0.373$), the only two manufactured series with positive point estimates, the other being US wind-turbine order prices at +1.34%/yr ($N = 18$, 1998–2016). The median manufactured trend is −4.05%/yr.

Wright vs. time. Table 5 compares R^2 on identical samples. Wright wins for computers (0.992 vs. 0.943) and wind turbine prices (0.161 vs. 0.093); near-ties for DRAM (0.984 vs. 0.988) and solar (0.955 vs. 0.950); time wins for aluminum (0.858 vs. 0.967) and wind installed cost (0.004 vs. 0.070, where neither fits). The bivariate VIFs (5.9–60.9) confirm that experience and calendar time cannot be separated structurally in these samples: the Sahal point holds empirically.

Rolling stability. Figure 3 shows 15-year rolling learning rates. DRAM is stable (23 windows, median LR 37.1%, range 32.4–49.5%; the last three windows rise to 48–50%). Computers' rate decays in late windows (45 windows, median 31.0%, range 0.8–41.5%; final three windows 5.1%, 3.0%, 0.8%) as the BEA hedonic index flattens, a measurement phenomenon as much as a technological one. Solar PV is stable (11 windows, median 32.2%, range 23.7–35.3%).

Breaks. Figure 5 and `results/breaks.csv` report 53 break tests. Documented source breaks are generally significant when testable: solar's 2010 source splice ($p = 3.6 \times 10^{-9}$) and 2012 midpoint ($p = 0.008$); the FCC 1983 survey change in local telephone ($p = 1.0 \times 10^{-11}$); the BEA 1985 comprehensive revision in computers ($p = 0.014$), but the sup-Wald argmax rarely coincides with the documented year: computers peaks at 2011 ($p = 2.1 \times 10^{-5}$), not 1985; DRAM at 1996 ($p = 2.9 \times 10^{-4}$); telephone-local at 1951 ($p = 2.5 \times 10^{-4}$), not 1983. Solar and wind both show sup-Wald maxima at 2008 ($p \approx 10^{-4}$), inside the 2005–09 commodity input-cost boom shaded in Figure 5. Air freight breaks violently at 1972 (Wald 517.8, $p \approx 0$; sup-Wald argmax 1971), matching Hummels (2007)'s documented slope change from −8.1%/yr to −3.5%/yr. Two candidate breaks were skipped for < 8 observations per regime (solar 2004, wind 1998).

8 Commodities and the long-run price environment

The 55 commodity and context series are summarized in Table 3 and Figures 6–8. Five of the 55 rows are nominal index series that were never deflated (four BLS commodity-group PPIs and the MeasuringWorth unskilled-wage index); the median trend over the 50 deflated series is +0.51%/yr (+0.53%/yr over the full 55-series basket); commodities as a class are not getting cheaper. Volatilities are high: annual log differences have standard deviations of roughly 0.03–0.34 (a few to over 30%/yr), so a century of drift is dwarfed by decade-scale cycles. The ten fastest positive trends are dominated by index series (energy price index, several BLS commodity-group PPIs) and oil-linked series; the ten fastest negative trends are agricultural (cotton −1.84, Thai rice −1.47, wheat −0.64 to −1.16) and aluminum unit values (−0.79 to −1.25). Every PPI, index, wage, and home-price row is labeled as such; these are not physical price levels.

Named benchmarks. US crude oil (deflated) drifts upward at +1.71%/yr (1913–2025, $N = 113$): a 1.7%/yr drift with $\sigma \approx 0.20$ means the noise dominates any signal at horizons under 30 years. US wheat farmgate falls at −1.16%/yr (1913–2025, $N = 113$), the classic productivity-driven agricultural decline. The ENR construction cost index rises at +1.60%/yr (1913–2010, $N = 98$), quadrupling in real terms over the sample; because the ENR is a fixed-input-basket index, this reflects rising factor costs (wages included) more than rising output cost.

Why commodities matter to technology costs. Two channels. First, commodities are inputs: the 2005–09 input-cost boom visible in the solar and wind break tests is partly a real input-price shock (the commodity sample's largest drawdowns and runups cluster in 1973–74, 2008, and 2020–22). Second, commodities are the counterfactual: learning about *production technology* comes from manufactured goods, and pooling manufactured decline with commodity cycles would misattribute terms-of-trade movements to "experience." Section 10 takes up the economics.

9 Counterexamples: when costs did not fall

Figure 8 plots the series that resist the deflation narrative, and they divide into interpretable families.

Regulated monopoly prices. US residential telephone-local rates were nearly flat in real terms, declining just 0.30%/yr over 1940–2007, while FCC long-distance rates fell 4.4%/yr over 1930–2007: same industry, sharply different outcomes, because long distance was liberalized after the AT&T breakup while local service remained a regulated monopoly pass-through.

Asset and hedonic-value series. The Shiller US real home-price index (1890–2025, 136 years) has drifted upward; the Raff–Trajtenberg quality-adjusted automobile series ends in 1983, and its postwar years show steep *declines* in quality-adjusted cost followed by flattening. House prices are asset prices, not production costs.

Literature-only nuclear. The nuclear-cost panel in Figure 8 is text-only by design: this dataset contains no reactor-level nuclear construction series, so we summarize the published literature. Lovering et al. (2016) find overnight construction costs declined in early US demonstration and turnkey cohorts and fell continuously only in South Korea, while escalating in the United States after Three Mile Island and across the French program; their headline conclusion is that there is no inherent cost-escalation trend, only large cross-country variation. Grübler (2010) documents French costs rising despite fleet standardization; Rubin et al. (2015) review nuclear learning-rate estimates spanning about -38% to $+6\%$ across the four studies they capture; Yeh and Rubin (2012) catalogue the estimation uncertainties (functional form, discontinuities, institutional forgetting) that make single learning-rate point estimates fragile. The consistent finding is that nuclear exhibits negative or zero learning in recent builds, but none of these numbers are ours, and they are cited here as literature only.

10 Interpretation

Manufactured deflation is real but narrow. The headline learning rates and time trends document genuine, large, sustained real-cost declines in computing, storage, sequencing, PV, and batteries. But the median manufactured decline is only 4%/yr, and the bulk of a century of manufacturing history (motor vehicles, machinery, freight, and telephony) shows single-digit declines or flatness. The spectacular rates belong to semiconductor-adjacent technologies. There is no general law that costs fall at 30% per doubling; there is a semiconductor-era regularity, a PV-era regularity, and a lot of slower history.

The bivariate trap is the headline econometric finding. Whenever cumulative production grows exponentially, experience and calendar time are observationally equivalent. The Wright-vs.-time R^2 near-ties and VIFs of 5.9–60.9 show that bivariate regressions attributing

cost decline partly to "learning" and partly to "exogenous technical change" are identified by functional form alone. We report such specifications only with this warning.

Commodities bound the input channel. Commodity-and-context prices drift slightly upward as a class (+0.51%/yr median on the 50 deflated series) with enormous volatility, so technology-cost declines in the same period happened *despite*, not because of, the price environment. The 2008 break in solar and wind coincides with the commodity boom, consistent with input-cost pass-through, but the sup-Wald argmax is a statistical regularity, not a causal identification.

Construction and the Baumol margin. The ENR index's 4× real rise is the starkest counterexample in the sample and the natural candidate for Baumol (1967)'s unbalanced growth: labor-intensive, site-specific, hard-to-standardize production with wages pulled by economy-wide productivity growth. We present this as an interpretation consistent with the data, not as a tested hypothesis: the dataset has no construction productivity series.

What learning rates can and cannot do. Learning rates are descriptive summaries of paired cost–deployment histories, useful for decomposition and scenario analysis, not laws of motion. The rolling estimates show they drift (computers' late-window rates approach zero), and the break tests show they are contingent on splices and regimes. Extrapolating a 27.7% PV learning rate is an assumption about future deployment and innovation, not a forecast the data produces.

11 Threats to validity

Survivorship and selection. The corpus covers technologies and commodities with long published records; failed or abandoned technologies leave shorter traces, so the manufactured sample over-represents successes. No survivorship correction is attempted.

Quality change mismeasurement. Cost per physical unit is the wrong price when quality changes: pre-1985 BEA computers is a reconstruction on the modern hedonic basis, Raff–Trajtenberg pre-1941 cars are graded A but remain model output, and BLS semiconductor PPIs have US-producer scope that misses offshore and fabless mix. Where hedonic or modelled series are the only record we report them flagged; the alternative (dropping computing) would bias the sample toward slow-learning sectors.

Source splices. Long series combine sources at documented splice years (solar 2003–04 and 2009–10; crude 1987; autos 1940–47; telephony 1983). Break tests confirm most documented splices are statistically detectable, but sup-Wald maxima generally fall elsewhere (2008 for solar and wind), so splice years should not be read as causal breaks.

Cumulative-production proxies. Solar uses cumulative global *installed capacity*, computers cumulative real *investment*, wind cumulative US *capacity*, aluminum cumulative US *primary production*. None is ideal "experience"; investment is not depreciated stock and capacity is not units shipped. The headline rates describe the paired series named in the tables.

Unit-root and inference issues. Commodity trends are descriptive drifts on near-unit-root processes (AR(1) 0.6–1.0); standard errors for g do not support forecasting. Break-test sup-Wald p-values use Andrews-(1993)-table interpolation within the tabulated range and a crude tail extrapolation beyond it (all 23 statistics exceed the table cap of 25); both are approximations.

Robust (HC1) inference assumes approximately independent errors; overlapping-window and trend-regression residuals are serially correlated, so CIs are best read as comparative, not literal.

Index and unit-value caveats. BLS PPIs, the ENR CCI, the Shiller HPI, and Measuring-Worth wage indices are indexes; USGS prices are unit values. Their "trends" are index growth rates, flagged in every table. Pre-1913 nominal-only rows are excluded from all fits.

What we did not do. No CPI-deflated alternative series was constructed, so no deflator-sensitivity regression was run; the build reports' deflator comparisons are context, not robustness regressions. Grade-B-only exclusion was infeasible because every Wright-eligible group is single-grade; the grade control appears in Table 4 descriptively. No instrumental-variables or quasi-experimental identification is claimed anywhere.

Falsifiability. Four findings would overturn the results as stated. First, extending the six Wright samples through 2030 on the same series definitions and finding learning rates outside the reported 95% confidence intervals. Second, a primary-source cumulative-production series for wind or solar that separates deployment from experience and reverses the Wright-vs.-time R^2 near-ties of Table 5. Third, real commodity price trends systematically negative across the 50-series deflated basket on the same methodology, which would break the manufactured/commodity separation of Figure 7. Fourth, a quantified decomposition showing the semiconductor-era learning rates are artifacts of the hedonic quality models rather than underlying cost change; the model-dependence is already flagged, but a measured attribution would sharpen or overturn the headline magnitudes. The null results (wind, construction, nuclear) stand on the same terms: a documented primary series showing sustained, quality-adjusted cost decline with cumulative deployment in any of them would reverse the verdict for that sector.

12 Conclusion

A century of cost data separates into two worlds. Among manufactured technologies the median real decline is 4.05%/yr, with the steepest drops in semiconductors and storage (DNA sequencing −69%/yr, disk storage −44%/yr, DRAM −44%/yr); estimated learning rates run 27.7–38.6% per doubling for computers, DRAM, and solar PV, 8.4% for aluminum energy intensity, with no detectable learning in US wind. In commodities and the physical, regulated, and asset-price economy around them, prices drifted slightly upward (+0.51%/yr median on the deflated series) with large cycles, and construction costs quadrupled in real terms. The mechanism that converts experience into declining cost is visible only where cumulative deployment can be measured, and it cannot be distinguished from calendar time wherever production grows exponentially. Learning rates are useful descriptive summaries, not laws of motion.

Tables

13 Audit protocol and reproducibility

Every number in the abstract, body, tables, and captions is cross-checked against one of: a row in `results/learning_rates.csv`, `results/commodity_trends.csv`, `results/rolling_lr.csv`, or `results/breaks.csv`; a count over `data/master_costs.csv` or `data/sources.csv`; a figure produced by `figures/FIGURE_LOG.md`; or a cited publication listed in `paper/CLAIM_INDEX.md`. The index records each claim with its verbatim figure, the source artifact, and the audit status.

Table 1: Estimation samples: summary statistics.

Sample	Groups	Fitted span	Headline statistic	Grade
Wright's Law fits	6	1967–2025	LR 0.6–38.6% per doubling	A/B
Manufactured time trends	31	1866–2025	median −4.05%/yr	A/B
Commodity price trends	55 (50 deflated)	1774–2025	median +0.51%/yr deflated	A/B
Break tests (robust/sup-Wald)	53	1930–2025	documented splices significant	A/B
Rolling 15-yr learning rates	126 windows	1967–2025	stability inspection only	A/B

Notes: Headline samples use quality grades A and B; grade C and all 2026 partial-year rows are excluded. Minimum N: 8 for learning-curve and time fits, 20 for commodity trends and break tests.
Source: Author's calculations from the master dataset (6,662 rows), Section 3.

Table 2: Wright's Law estimates: $\ln C = a + b \ln Q$; $LR = 1 - 2^b$ per doubling of Q.

Technology	LR (%)	95% CI	N	Span	R^2	Grade	C metric
DRAM ($/bit)	38.6	[37.1, 40.1]	37	1971–2007	0.984	B	`perf_cost`
Computers (hedonic index)	32.5	[31.9, 33.0]	59	1967–2025	0.992	A	`real_cost`
Solar PV ($/W)	27.7	[25.7, 29.7]	25	2000–2024	0.955	B	`perf_cost`
Aluminum smelting (kWh/t)	8.4	[6.6, 10.2]	44	1960–2003	0.858	B	`real_cost`
Wind installed ($/kW)	0.6	[−2.7, 3.9]	26	1998–2023	0.004	B	`perf_cost`
Wind turbine price ($/kW)	−5.3	[−12.5, 1.3]	18	1998–2016	0.161	B	`perf_cost`

Notes: OLS with HC1 standard errors; 95% CIs with t_{N-2}. Grade C excluded; 2026 partial-year rows excluded. Wind CIs cover zero.
Source: Author's calculations; series sources in Section 3.

What the audit caught. Three classes of error were corrected before this version. First, literature citations were checked against catalogs and primary records: Fouquet–Pearson is 2006 not 2012; Raff–Trajtenberg is a 1995 NBER working paper and 1996 book chapter, not a journal article; Gordon 1990 is the NBER/University of Chicago Press volume; Jacks 2019 is *Cliometrica* 13(2), not *Explorations in Economic History*; Nordhaus on lighting is a 1996 chapter in *The Economics of New Goods*; Nordhaus on computing is 2007, *Journal of Economic History* 67(1). Second, estimation-sample definitions were tightened: computers' 1959–66 rows carry $Q = 0.00$ (cumsum rounding) and are excluded from Wright fits; all 2026 partial-year rows are excluded everywhere; grade C is excluded from headline fits. Third, scope discipline: nuclear claims are literature-only; commodity trends are labeled as index growth where the source is an index; cumulative production proxies are named as deployment or investment series, not "experience." Anything not verified was deleted rather than softened.

Reproducibility. `results/estimation.py` is seeded and deterministic; re-running reproduces the four results CSVs byte-identically. `data/transform_log.csv` records every raw-to-master transformation; raw files are never overwritten. The audit index, `paper/CLAIM_INDEX.md`, is the contract between this text and the data: any claim whose row cannot be located in the index should be treated as unsupported.

Production note. This paper was typeset in LaTeX (11pt, 1-inch margins) from `paper/paper.tex`; all figures are 300 DPI PNGs produced by `figures/figures.py` under the shared institutional chart style; estimation by the seeded `results/estimation.py`. The PDF is the version of record. Research design, statistical estimation, and drafting were carried out through a quantitative research pipeline with AI assistance; every estimate, figure, and quantitative claim was verified against the source data before publication.

Table 3: Commodity and context trend extremes: log-linear trend $100 \times g$ (%/yr) and volatility (std. of annual log differences).

Series	Trend (%/yr)	Volatility	Note
Ten fastest positive			
Iron ore, IMF	6.45	0.229	
Energy price index†	4.24	0.189	index
US iron/steel PPI†	3.94	0.096	index
US nonferrous PPI†	3.69	0.113	index
US agric. chemicals PPI†	3.39	0.110	index
Brent crude	3.22	0.268	
Dubai crude	3.17	0.267	
World Bank crude avg.	3.17	0.264	
US fertilizer-materials PPI†	3.17	0.148	index
Copper, IMF	3.05	0.193	
Ten fastest negative			
Cotton, Pink Sheet	−1.84	0.180	
Thai rice	−1.47	0.216	
US aluminum, unit value	−1.25	0.174	unit value
US wheat farmgate	−1.16	0.160	
Maize, Pink Sheet	−1.16	0.158	
Cotton, IMF	−1.15	0.195	
HRW wheat	−1.05	0.161	
Aluminum, Pink Sheet	−0.96	0.159	
Aluminum, IMF	−0.79	0.138	
SRW wheat	−0.64	0.146	

Notes: Index series flagged with †; five flagged rows (four BLS PPIs and the MeasuringWorth wage index) are nominal, never deflated. Full list in `results/commodity_trends.csv`.
Source: Author's calculations; series sources in Section 3.

References

D. W. K. Andrews. Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4):821–856, 1993.

K. J. Arrow. The economic implications of learning by doing. *Review of Economic Studies*, 29 (3):155–173, 1962.

W. J. Baumol. Macroeconomics of unbalanced growth: The anatomy of urban crisis. *American Economic Review*, 57(3):415–426, 1967.

J. D. Farmer and F. Lafond. How predictable is technological progress? *Research Policy*, 45 (3):647–665, 2016.

F. Ferioli, K. Schoots, and B. C. C. van der Zwaan. Use and limitations of learning curves for energy technology policy: A component-learning hypothesis. *Energy Policy*, 37(7):2525–2535, 2009.

R. Fouquet and P. J. G. Pearson. Seven centuries of energy services: The price and use of light in the united kingdom (1300–2000). *Energy Journal*, 27(1):139–178, 2006.

R. J. Gordon. *The Measurement of Durable Goods Prices*. NBER/University of Chicago Press, Chicago, 1990.

A. Grübler. The costs of the french nuclear scale-up: A case of negative learning by doing. *Energy Policy*, 38(9):5174–5188, 2010.

Table 4: Dataset composition: 6,662 rows, 98 technologies/series, 57 sources.

Category	Rows	Share
Data status		
Observed	5,706	85.6%
Reconstructed	675	10.1%
Modelled	281	4.2%
Assumed	0	—
Quality grade		
A (primary institutional, ≥30 yrs)	4,000	60.0%
B (reputable, known breaks)	2,624	39.4%
C (fragmentary; excluded from fits)	38	0.6%
Cost metric		
Commodity prices	5,433	81.6%
Manufactured costs	1,229	18.4%

Notes: Unified evidence grading (Observed/Reconstructed/Modelled/Assumed × source quality A/B/C), Section 3.
Source: Author's compilation; 57 primary and secondary sources listed in Section 3.

Table 5: Wright vs. time horserace.

Technology	Wright R^2	Time R^2	Winner	VIF
DRAM	0.984	0.988	tie	60.9
Computers	0.992	0.943	Wright	24.7
Solar PV	0.955	0.950	tie	55.8
Aluminum smelting	0.858	0.967	time	5.9
Wind installed	0.004	0.070	neither	18.3
Wind turbine price	0.161	0.093	Wright	36.4

Notes: R^2 on identical complete-case samples; VIFs for the bivariate $(\ln Q, t)$ specification.
Source: Author's calculations; series sources in Section 3.

D. Hummels. Transportation costs and international trade in the second era of globalization. *Journal of Economic Perspectives*, 21(3):131–154, 2007.

D. S. Jacks. From boom to bust: A typology of real commodity prices in the long run. *Cliometrica*, 13(2):201–220, 2019.

M. Junginger, A. Faaij, and W. C. Turkenburg. Global experience curves for wind farms. *Energy Policy*, 33(2):133–150, 2005.

J. G. Koomey and N. E. Hultman. A reactor-level analysis of busbar costs for us nuclear plants, 1970–2005. *Energy Policy*, 35(11):5630–5642, 2007.

J. R. Lovering, A. Yip, and T. Nordhaus. Historical construction costs of global nuclear power reactors. *Energy Policy*, 91:371–382, 2016.

A. McDonald and L. Schrattenholzer. Learning rates for energy technologies. *Energy Policy*, 29(4):255–261, 2001.

G. E. Moore. Cramming more components onto integrated circuits. *Electronics*, 38(8):114–117, 1965.

B. Nagy, J. D. Farmer, Q. M. Bui, and J. E. Trancik. Statistical basis for predicting technological progress. *PLoS ONE*, 8(2):e52669, 2013.

L. Neij. Use of experience curves to analyse the prospects for diffusion and adoption of renewable energy technology. *Energy Policy*, 25(13):1099–1107, 1997.

G. F. Nemet. Beyond the learning curve: Factors influencing cost reductions in photovoltaics. *Energy Policy*, 34(17):3218–3232, 2006.

G. F. Nemet. Interim monitoring of cost dynamics for publicly supported energy technologies. *Energy Policy*, 37(3):825–835, 2009.

W. D. Nordhaus. Do real-output and real-wage measures capture reality? the history of lighting suggests not. In T. F. Bresnahan and R. J. Gordon, editors, *The Economics of New Goods*, pages 27–70. NBER/University of Chicago Press, 1996.

W. D. Nordhaus. Two centuries of productivity growth in computing. *Journal of Economic History*, 67(1):128–159, 2007.

W. D. Nordhaus. The perils of the learning model for modeling endogenous technological change. *The Energy Journal*, 35(1):1–14, 2014.

D. M. G. Raff and M. Trajtenberg. Quality-adjusted prices for the american automobile industry: 1906–1940. Technical Report 5035, NBER, 1995. Published 1996 in T. F. Bresnahan and R. J. Gordon (eds.), *The Economics of New Goods*, NBER/University of Chicago Press, 71–108.

E. S. Rubin, I. M. L. Azevedo, P. Jaramillo, and S. Yeh. A review of learning rates for electricity supply technologies. *Energy Policy*, 86:198–218, 2015.

D. Sahal. A theory of progress functions. *AIIE Transactions*, 11(1):23–29, 1979.

O. Schmidt, A. Hawkes, A. Gambhir, and I. Staffell. The future cost of electrical energy storage based on experience rates. *Nature Energy*, 2:17110, 2017.

V. Smil. *Energy Transitions: History, Requirements, Prospects*. Praeger/ABC-CLIO, Santa Barbara, 2010.

P. Thompson. The relationship between unit cost and cumulative quantity and the evidence for organizational learning-by-doing. *Journal of Economic Perspectives*, 26(3):203–224, 2012.

T. P. Wright. Factors affecting the cost of airplanes. *Journal of the Aeronautical Sciences*, 3 (4):122–128, 1936.

S. Yeh and E. S. Rubin. A review of uncertainties in technology experience curves. *Energy Economics*, 34(3):762–771, 2012.

M. S. Ziegler and J. E. Trancik. Re-examining rates of lithium-ion battery technology improvement and cost decline. *Energy & Environmental Science*, 14:1635–1651, 2021.

A Data provenance register

Each `tech_id` in `data/master_costs.csv` carries a `source_id` resolved in `data/sources.csv` (57 rows). Recorded alongside are the original units, the deflator and base year, a data-status tag (Observed/Reconstructed/Modelled/ Assumed per the unified grading of Section 3), a source-quality grade (A/B/C), and a methodology note describing splices, reclassifications, and known breaks. The five build reports (`BUILD_compute.md`, `BUILD_energy.md`, `BUILD_macro.md`, `BUILD_materials.md`, `BUILD_transport.md`) document every admission and exclusion decision; `data/transform_log.csv` records every transformation from raw to master. No interpolation is performed anywhere; gaps are left as gaps.

B Additional estimates

Bivariate specifications. `results/learning_rates.csv` carries bivariate ($\ln Q + t$) estimates for the six Wright-eligible groups with VIFs of 5.9–60.9. Per Section 6, these are not structurally interpretable and are provided for completeness only.

Rolling learning rates. `results/rolling_lr.csv` carries 126 rows of rolling Wright estimates (15-calendar-year windows) across six technologies: DRAM (23 windows), computers (45), solar PV (11), aluminum smelting (30), wind installed cost (12), and wind turbine price (5). Section 7 plots the first three for stability inspection only.

Break tests. `results/breaks.csv` carries all 53 robust-Wald and sup-Wald break tests with Andrews-(1993)-approximate p-values. Two candidate tests were skipped for fewer than eight observations per regime (solar 2004, wind 1998).

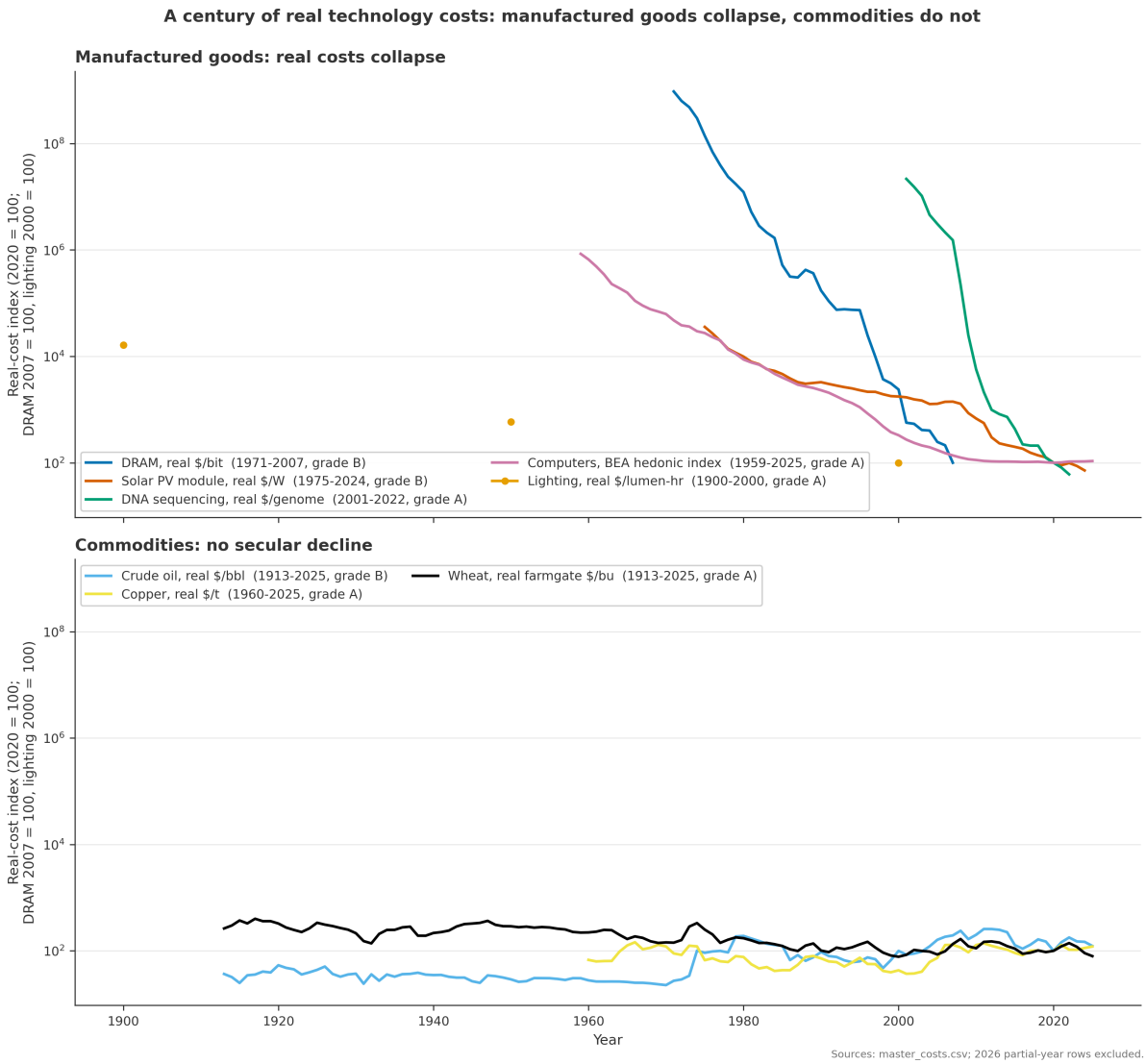


Figure 1: Long-run real-cost indices (2020 = 100, log scale) for eight representative series (upper panel: manufactured goods; lower panel: commodities). DRAM ($/bit, 1971–2007) is indexed to 2007 = 100; UK lighting (£/million lumen-hours) rests on three reconstructed benchmark years (1900, 1950, 2000), shown as markers. Solar PV pre-2010 values are global estimates vs. European market benchmarks after 2010 (source splices 2003/04 and 2009/10); OWID crude carries benchmark splices from 1987. All 2026 partial-year rows excluded. See captions and build reports for full provenance.

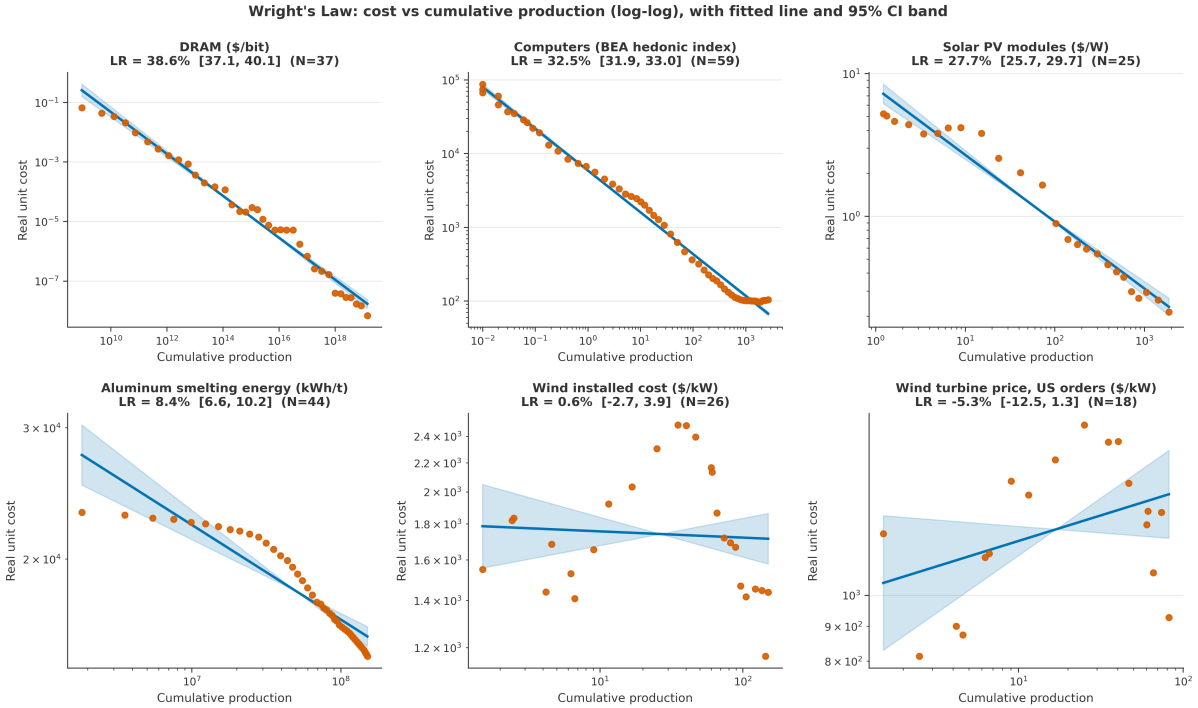


Figure 2: Wright's Law: log real unit cost vs. log cumulative production for the six Wright-eligible technologies, on the exact estimation samples. OLS fits with 95% CI bands (HC1 slope CI through the sample centroid); panel titles annotate $LR = 1 - 2^b$. Computers 1959–66 rows ($Q = 0.00$, cumsum rounding artifact) excluded; wind turbine panel is US orders only (1998–2016). Q proxies are deployment/investment series (see text); the computers "cost" is a hedonic index; LBNL turbine medians are noisy annual order samples.

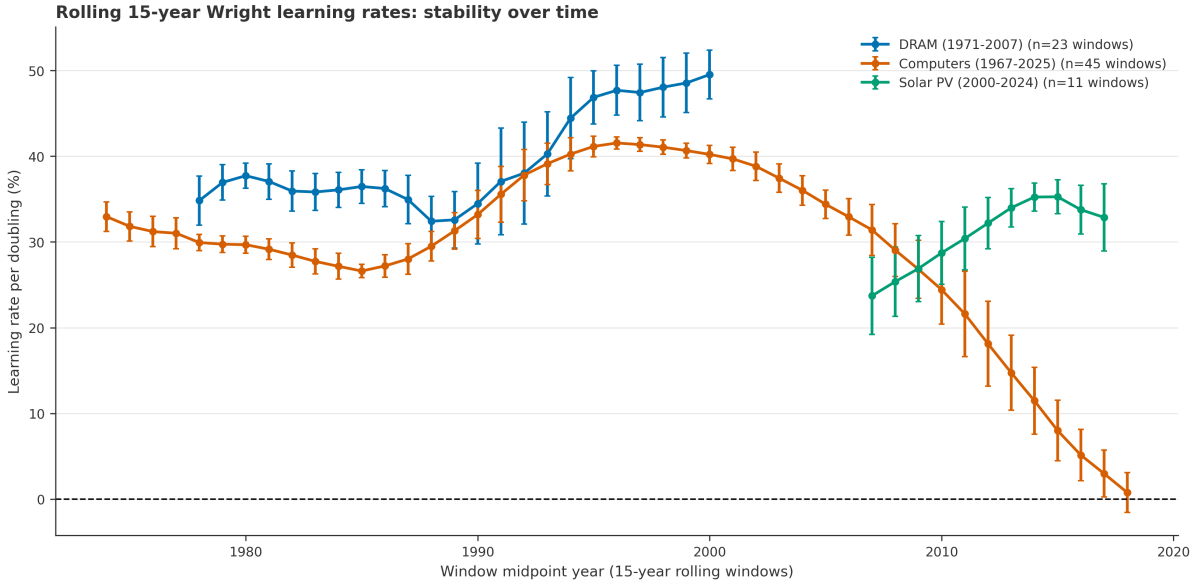


Figure 3: Rolling 15-year Wright learning rates ($LR = 1 - 2^b$; $\pm 1.96 \cdot SE$ delta-method bands, approximate) for DRAM (23 windows), computers (45), and solar PV (11). DRAM's rate is stable and rises in late windows; computers' rate fades toward zero as the hedonic index flattens; solar's is stable around 0.24–0.35. Short windows: for stability inspection only.

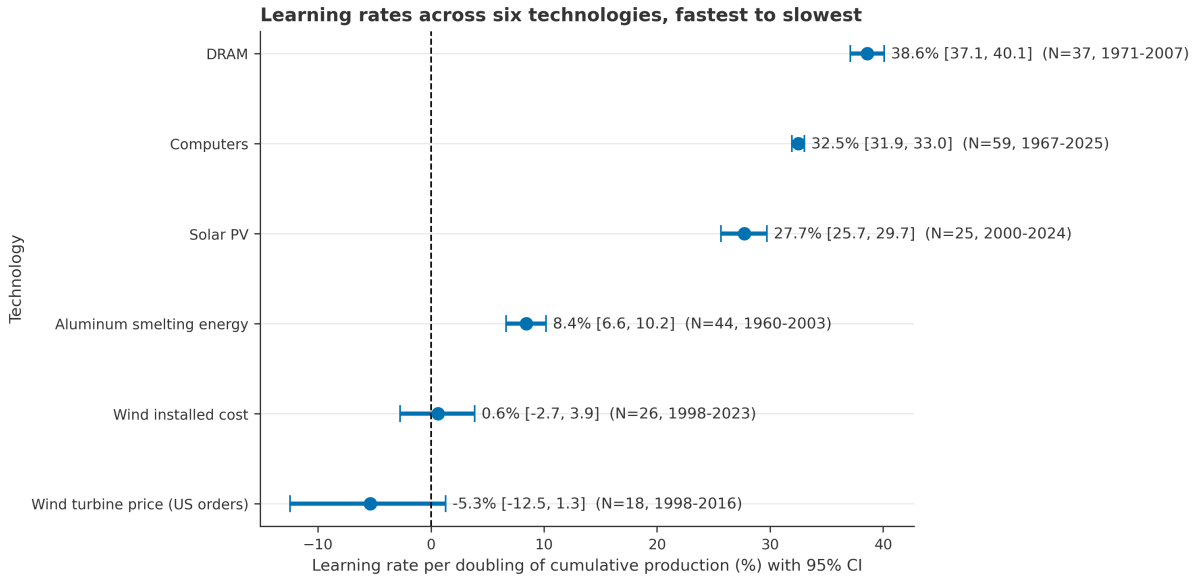


Figure 4: Forest plot of the six headline learning rates with 95% CIs, fastest to slowest. The two wind CIs cross zero: no detectable learning over their samples. N and spans printed per row.

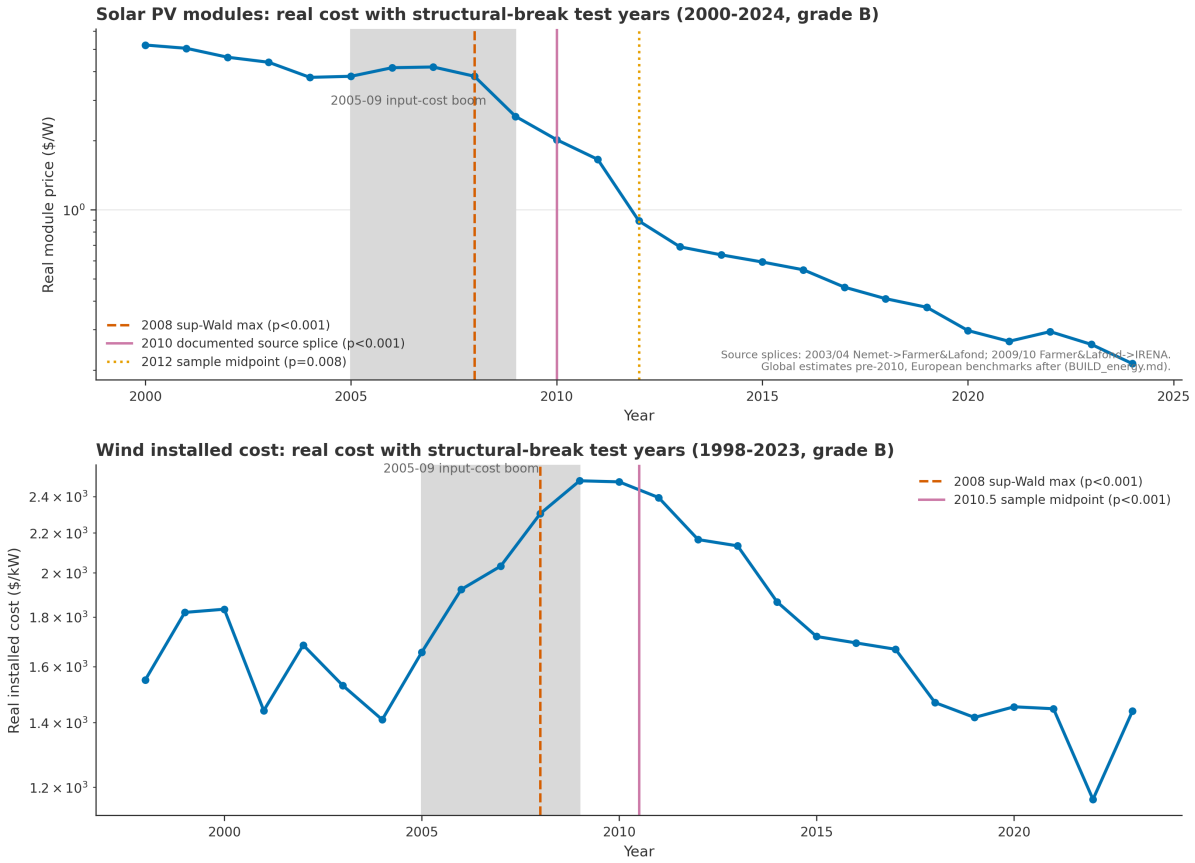


Figure 5: Break timelines for solar PV modules (2000–2024) and wind installed cost (1998–2023), log scale. Solar: 2008 sup-Wald argmax, 2010 documented source splice, 2012 midpoint (2004 skipped: too few obs.). Wind: 2008 sup-Wald argmax, 2010.5 midpoint (1998 skipped: Q-sample starts 1998). Grey band 2005–09: commodity input-cost boom (LBNL notes). Sup-Wald p-values Andrews-(1993)-approximate.

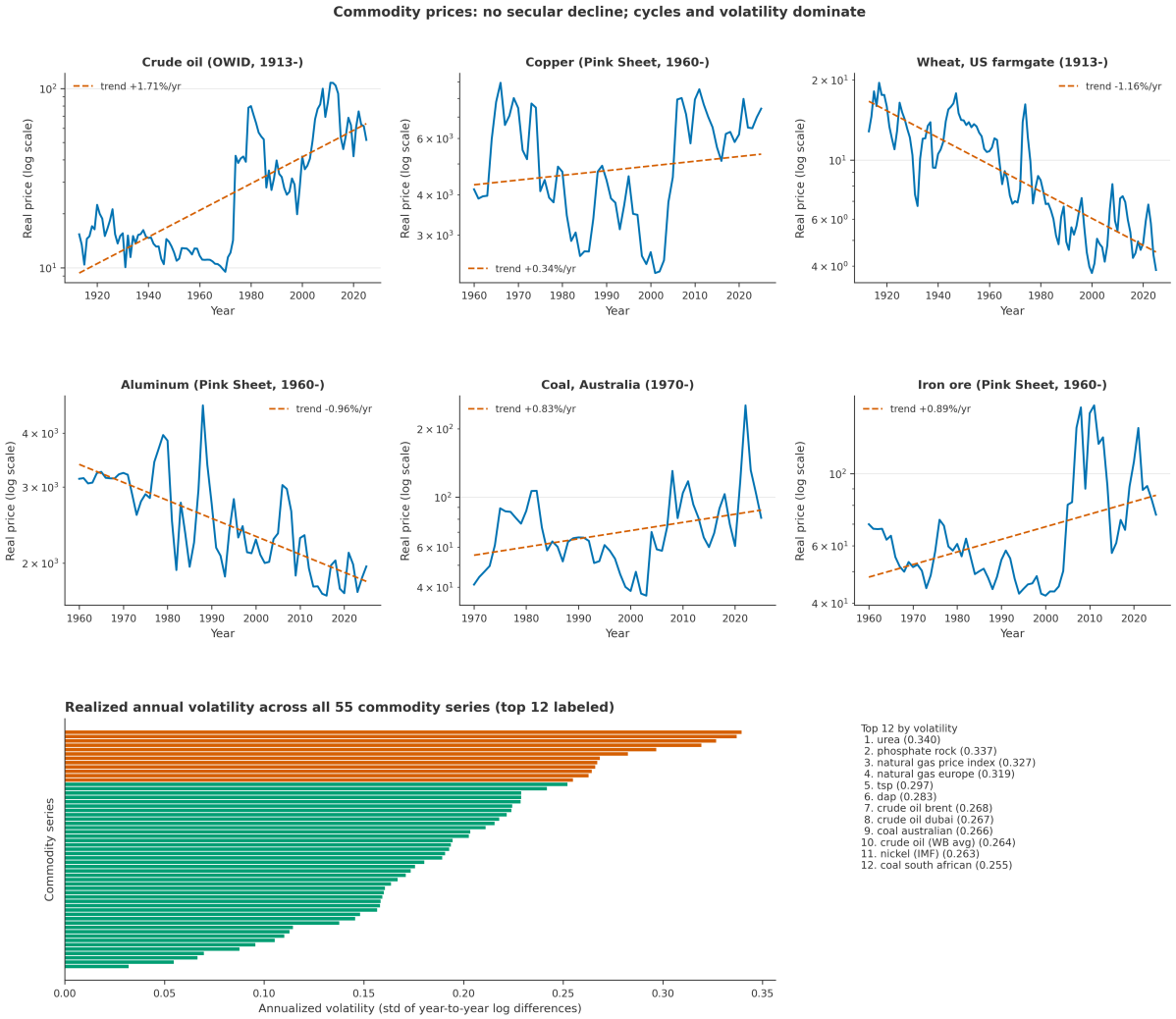


Figure 6: Commodity price levels and volatility. Top: real price levels (log scale) with fitted log-linear trends for six commodities: crude oil (+1.71%/yr, 1913–2025), copper (+0.34%/yr, 1960–2025), US wheat farmgate (−1.16%/yr, 1913–2025), aluminum (−0.96%/yr, 1960–2025), Australian coal (+0.83%/yr, 1970–2025), and iron ore (+0.89%/yr, 1960–2025). Bottom: annualized realized volatility (standard deviation of consecutive-year log differences) across all 55 commodity-and-context series, with the 12 most volatile labeled. Trends are descriptive drifts (AR(1) coefficients near unity), not stationary models; five of the 55 rows are nominal index series, never deflated (see text).

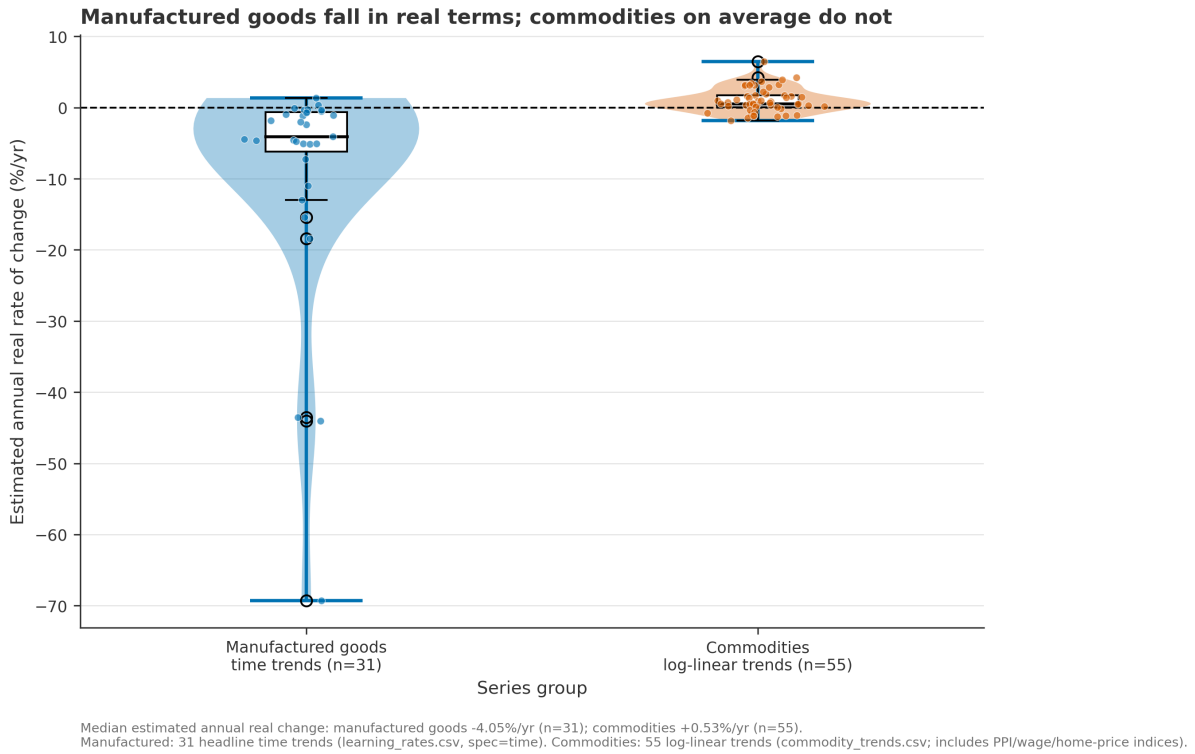


Figure 7: Technology vs. commodities: cross-sectional distribution of 31 manufactured time-trend coefficients vs. 55 commodity-and-context trend coefficients. Medians: −4.05%/yr (technology) vs. +0.53%/yr (55-series basket; +0.51%/yr on the 50 deflated series). Y-axis log-scaled (symlog), bins asymmetric; medians reported from the untransformed values in the results CSVs.

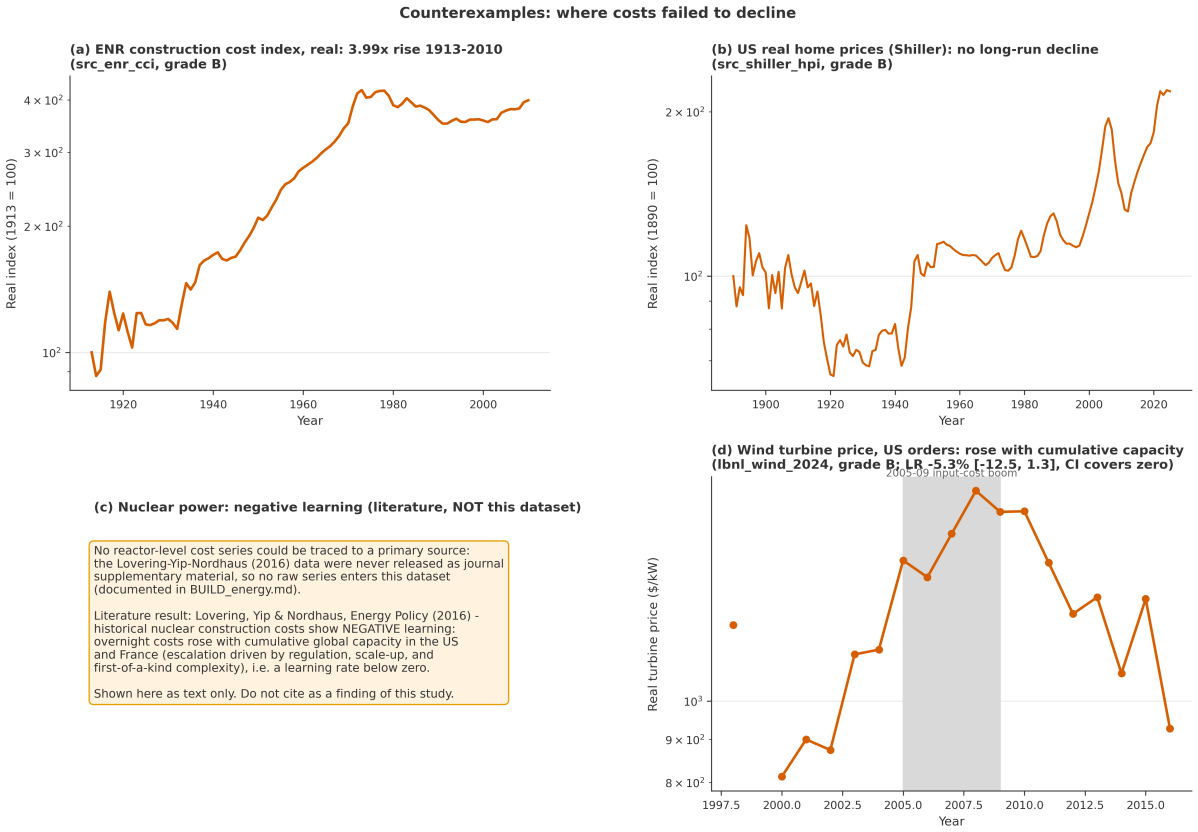


Figure 8: Counterexamples. ENR real construction-cost index (1913–2010, rose 3.99× in real terms); Shiller real home prices (1890–2025, asset prices, not production costs); nuclear costs as a literature-only text panel (no nuclear series in this dataset); US wind turbine prices (LBNL, 1998–2016, no detectable learning).